

Why AlphaFold

- Obtaining atomic-resolution protein structures experimentally is time and resource intensive.
- Google DeepMind's AlphaFold has emerged as a breakthrough AI / ML tool, capable of predicting protein structures at atomic resolution without relying on known structural templates.
- An example of how AI/ML can directly benefit scientific research and human health.

Motivation

- Many potential AlphaFold users do not have access to HPC environments or are unfamiliar with Linux terminals, command-line interfaces (CIL), and shell scripting.
- A user-friendly "open-source" GUI would lower the barrier for those users to entry for running AlphaFold.

SigmaFold: Open-Source AlphaFold WebApp

- Integrates AlphaFold2 workflow and resources developed by OSC with a browser-based interface hosted in OSC.
- Employ HTML, CSS, JavaScript, Flash, Python, Bash, Slurm. (<http://GitHub.com/Lumonike/SigmaFold>)
- Users input protein ID.
- Automates the prediction process and provides:
 - Visualize 3D protein structure using 3Dmol.js
 - Create pLDDT and PAE plots to determine model accuracy.

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AlphaFold2 Workflow

